Module 03a — OOP Part 1: Classes & Encapsulation

Phase: Advanced Python
Previous module: Modules and Packages
Next module: OOP Part 2 — Inheritance & Composition

Series: OOP overview · Part 1 (this file) · Part 2 · Part 3 · Part 4

Assignment (take-home): The portfolio **Mini-Project (Assignment)** is in Part 3 — BankAccount OOP CLI. This Part 1 file is class + lab drills; start the lab kit here and push in Part 3.

Lab kit: 15_Lab_Assets/17_bank_account_oop/

Motivation — Feel the pain of dicts + free functions

Look at this student tracker. Data lives in a dict; behaviour lives in separate functions:

```
s1 = {"roll": 101, "name": "Anita", "marks": [72, 68, 81]}
s2 = {"roll": 102, "name": "Vikram", "marks": [40, 35, 48]}

def average(student):
    marks = student["marks"]
    return sum(marks) / len(marks)

def grade(student):
    avg = average(student)
    if avg >= 85:
        return "A"
    if avg >= 70:
        return "B"
    if avg >= 50:
        return "C"
    return "Fail"

def display(student):
    print(f"{student['roll']}: {student['name']} — {grade(student)}")

display(s1)
display(s2)

# Easy mistakes in a busy script:
s1["marks"] = -10          # nonsense value, no guard
del s2["name"]             # KeyError waiting to happen
print(average({"marks": []}))  # ZeroDivisionError — not really a student
```

Expected Output (happy path only):

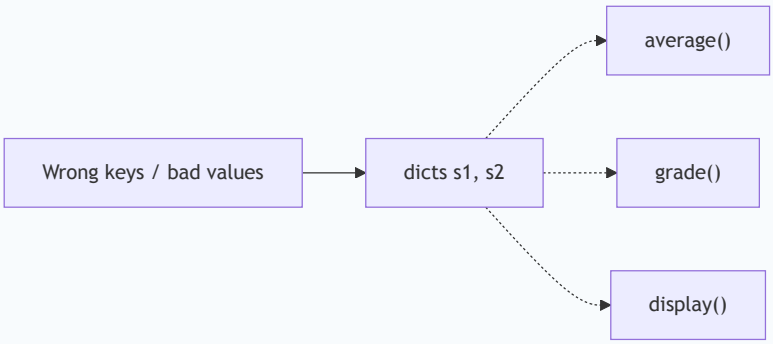
```
101: Anita — B
102: Vikram — Fail
```

Problems with this style:

1. Anyone can corrupt the dict (`marks = -10`, delete `name`).
2. Related data and behaviour are **scattered** — easy to pass the wrong shape.

3. Adding `college`, `email`, or `is_pass()` means hunting every function that touches a student.
4. As the program grows, the same broken pattern repeats for Order, Account, Product — harder to test and reuse.

Visual — data here, logic there (easy to break):



Key takeaways:

- Dicts hold data; free functions hold behaviour — easy to desync
- Callers can mutate or delete keys freely
- Same pain repeats for Order / Account / Product as the app grows

Bundle **data + behaviour** into one unit — a **class**. Each filled form is an **object**:

```
class Student:
    def __init__(self, roll, name, marks):
        self.roll = roll
        self.name = name
        self.marks = list(marks)   # own copy — don't share the caller's list

    def average(self):
        if not self.marks:
            return 0.0
        return sum(self.marks) / len(self.marks)

    def grade(self):
        avg = self.average()
        if avg >= 85:
            return "A"
        if avg >= 70:
            return "B"
        if avg >= 50:
            return "C"
        return "Fail"

    def display(self):
        print(f"{self.roll}: {self.name} — {self.grade()}")

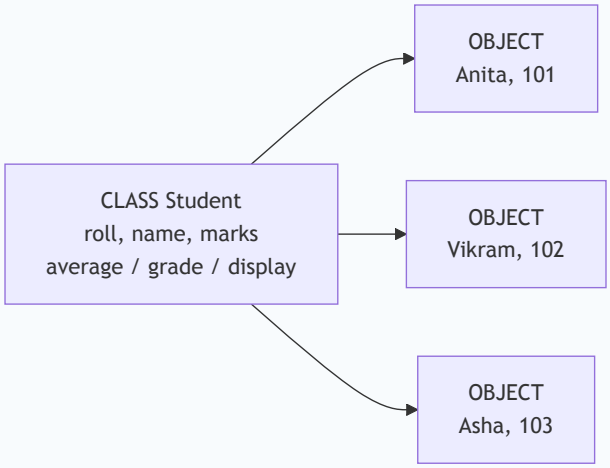
s1 = Student(101, "Anita", [72, 68, 81])
s2 = Student(102, "Vikram", [40, 35, 48])
s1.display()
s2.display()
print("Anita average:", round(s1.average(), 1))
```

Expected Output:

```
101: Anita — B
102: Vikram — Fail
Anita average: 73.7
```

That shift — from “bags of data + floating functions” to “objects that know how to behave” — is **Object-Oriented Programming (OOP)**.

Visual — one blueprint, many filled forms:



Key takeaways:

- **Class** = blueprint (form template)
- **Object** = one filled form (Anita, Vikram, …)
- Methods live **with** the data — call `s1.display()`, not `display(s1)` with a loose dict

What is OOP? / Why do we need it?

Object-Oriented Programming models software around **things** in the domain — a `Student`, a `BankAccount`, an `Order`.

🏦 Bank 📱 UPI 👤 Employee 💻 Laptop 📦 Order

Where used?

- **SecureBank** (accounts, UPI wallets)
- **TechMart** (orders, products, carts)
- **MedCare** (patients, appointments)
- **AcmeHR** (employees, salary slips)
- CRM (customers, tickets)

Idea	Meaning
`class`	Blueprint (like an Aadhaar form template)
`object`	One filled form — Rahul, roll 101
Attribute	Data on the object (`name`, `balance`)
`method`	Behaviour on the object (`deposit()`, `get_grade()`)

Without OOP (dicts + functions)	With OOP (classes)
Data and rules in different places	One type owns both
Easy to pass the wrong shape	Constructor sets a valid starting state
Hard to reuse across scripts	Import the class; create many objects
Growing apps become “dict spaghetti”	Clear nouns: Account, Order, Employee

Classic pillars (Part 1 focuses on encapsulation; Parts 2–3 cover inheritance and polymorphism):

Pillar	Plain English	Python taste
Encapsulation	Bundle data + methods; **hide** state; force changes through a validating API	Exact problem: stop `acct.balance = -50000` bypassing `withdraw` — use `_`/`__` + methods/`@property`
Abstraction	Show essentials; hide messy internals	Callers use `deposit()` / `withdraw()` without knowing balance rules; **ABC / interfaces** → Part 4
`inheritance`	Child reuses/extends parent (**IS-A**)	→ Part 2
Polymorphism	Same call, different behaviour	→ Part 3

Composition (HAS-A) is previewed here and practised in Part 2: an `Employee` **has a** `Laptop`.

Important Concepts

1. Class, object, `self`, `__init__`

Real-world analogy: An **ATM card slot** is the class — one machine design. Priya's debit card is one **object**; Ravi's card is another. Same fields (`holder`, `balance`), different values. `__init__` is the branch printing *this* card with *this* account number. `self` is "the card in the slot right now." **SecureBank** mints millions of `Account` objects from one class. **TechMart** does the same for `Order`. **MedCare** does the same for `Patient`.



Where used? Any entity you create many of: students, products, tickets, employees, accounts.

Definition: A **class** is the blueprint (data + behaviour together). An **object** is one concrete instance of that class. `__init__` runs when you create an object and sets starting data. `self` is the current object (like `this` in Java — but Python makes you write it).

Purpose: Real systems are easier as objects than as loose dicts passed through many free functions.

```
class Student:
    def __init__(self, roll, name):   # constructor
        self.roll = roll              # instance attribute
        self.name = name

    def display(self):                # instance method
        print(f"{self.roll}: {self.name}")

s1 = Student(101, "Anita")            # object creation
s2 = Student(102, "Vikram")
s1.display()
s2.display()
print("type(s1):", type(s1).__name__)
print("same class?", type(s1) is type(s2))
```

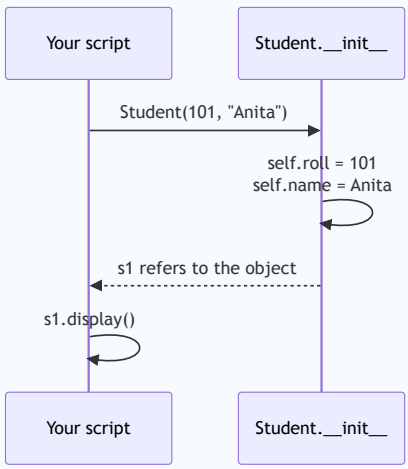
Expected Output:

```
101: Anita
102: Vikram
type(s1): Student
same class? True
```

What happens on `Student(101, "Anita")`?

1. Python creates a new empty object.
2. `__init__(self, 101, "Anita")` runs — `self` is that new object.
3. Attributes are attached to `self`.
4. The name `s1` now **refers** to that object in memory.

Visual — call → `__init__` fills the object → you use the reference:



Key takeaways:

- `ClassName(...)` creates an object and runs `__init__`
- `self` = the **current** object (passed automatically on `s1.display()`)
- Many objects can share one class; each has its own attribute values

Difference between Method and Constructor:

	Method	Constructor (`__init__`)
Name	Any valid name	Always `__init__`
When it runs	When you call it	Automatically on `ClassName(...)`
How often per object	As many times as you call	Once (at creation)
Typical job	Business logic	Declare and initialize instance data
Return	Optional value	Should not return a useful value

Remember: `self` = the **current object**. First parameter of every instance method. On `s1.display()`, Python secretly passes `s1` as `self`.

Common Error: forgetting `self` in the method signature → `TypeError: display() takes 0 positional arguments but 1 was given` when you call `s1.display()`.

Note:

- `s1` is a **reference** to the object — you use it to call methods and read attributes.

- `__init__` runs automatically after object creation — you rarely call it by name.
- If you omit `__init__`, Python still creates the object (empty shell); you can add attributes later (usually not ideal for real entities).
- Over-engineering OOP (five classes for hello world) — use functions until complexity grows.

Try these (3 project examples):

Program 1 — SecureBank savings account (deposit + mini statement)

```
class SavingsAccount:
    def __init__(self, holder, account_no, balance=0):
        self.holder = holder
        self.account_no = account_no
        self.balance = float(balance)

    def deposit(self, amount):
        if amount <= 0:
            print("Deposit must be positive")
            return
        self.balance += amount

    def statement(self):
        print(f"{self.account_no} | {self.holder} | ₹{self.balance:.0f}")

priya = SavingsAccount("Priya Shah", "SB-1001", 5000)
ravi = SavingsAccount("Ravi Menon", "SB-1002", 1200)
priya.deposit(1500)
ravi.deposit(-10)  # rejected
priya.statement()
ravi.statement()
```

Expected Output:

```
Deposit must be positive
SB-1001 | Priya Shah | ₹6500
SB-1002 | Ravi Menon | ₹1200
```

Program 2 — TechMart checkout line (SKU, qty, line total)

```
class OrderLine:
    def __init__(self, sku, name, unit_price, qty):
        self.sku = sku
        self.name = name
        self.unit_price = float(unit_price)
        self.qty = int(qty)

    def line_total(self):
        return self.unit_price * self.qty

    def label(self):
        return f"{self.sku} {self.name} x{self.qty} = ₹{self.line_total():.0f}"

cart = [
    OrderLine("NB-A4", "Notebook", 40, 3),
    OrderLine("PEN-01", "Gel pen", 10, 5),
]
for line in cart:
```

```
    print(line.label())
print("Cart total:", sum(line.line_total() for line in cart))
```

Expected Output:

```
NB-A4 Notebook x3 = ₹120
PEN-01 Gel pen x5 = ₹50
Cart total: 170
```

Program 3 — MedCare appointment slip (patient, doctor, slot)

```
class Appointment:
    def __init__(self, patient, doctor, slot):
        self.patient = patient
        self.doctor = doctor
        self.slot = slot
        self.status = "scheduled"

    def confirm(self):
        self.status = "confirmed"

    def slip(self):
        print(f"{self.patient} → Dr {self.doctor} @ {self.slot} [{self.status}]")

a1 = Appointment("Anita Sharma", "Meera Iyer", "10:30")
a2 = Appointment("Vikram Rao", "Kiran Das", "11:15")
a1.confirm()
a1.slip()
a2.slip()
```

Expected Output:

```
Anita Sharma → Dr Meera Iyer @ 10:30 [confirmed]
Vikram Rao → Dr Kiran Das @ 11:15 [scheduled]
```

---

2. Three kinds of variables (instance / class / local)

Real-world analogy: A **college ID card**. The college name is printed the same on every card — that is a **class** variable (one shared value). Your roll number is only yours — **instance**. The ink drying while the printer runs is **local**: it exists during that print job and disappears. **AcmeHR** shares `company` across employees. **SecureBank** shares `IFSC` across accounts at one branch. **TechMart** must *not* share one cart list across every customer (that is the mutable-class trap below).



Where used? Shared company / college / tax rate (class); per-person id and name (instance); temporary totals inside a method (local).

Definition:

- **Instance** — unique per object (`self.emp_id`)
- **Class** (sometimes called “static” in older notes) — one copy shared by all objects (`Employee.company`)
- **Local** — temporary names inside a method; die when the method ends

Purpose: Company name can be shared; employee id must not. Temporary totals should not leak as attributes.

```
class Employee:
    company = "KSR Datavizon"   # class attribute — shared

    def __init__(self, emp_id, name):
        self.emp_id = emp_id    # instance attribute — unique per object
        self.name = name

    def label(self):
        title = "Staff"         # local variable — method only
        return f"{title}: {self.name} @ {Employee.company}"

e1 = Employee(1, "Ravi")
e2 = Employee(2, "Meera")
print(e1.label())
print(e2.label())
print("company via class:", Employee.company)
```

Expected Output:

```
Staff: Ravi @ KSR Datavizon
Staff: Meera @ KSR Datavizon
company via class: KSR Datavizon
```

Instance vs class — the shadowing trap:

```
e1 = Employee(1, "Ravi")
e2 = Employee(2, "Meera")
e1.company = "Acme"   # creates an *instance* attribute on e1 only

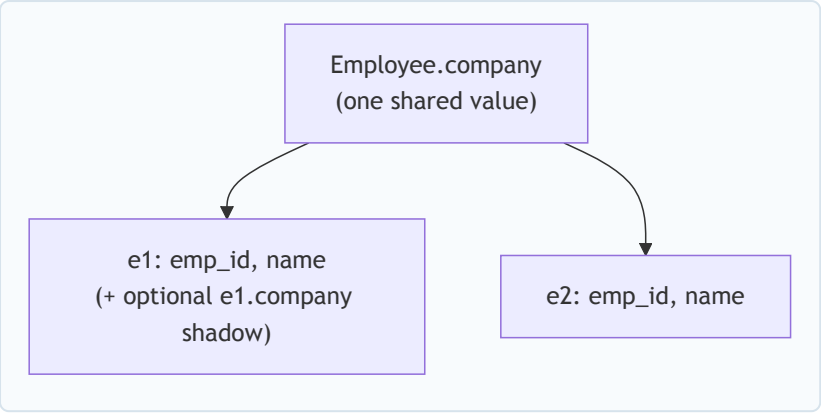
print(e1.company)         # Acme
print(e2.company)         # KSR Datavizon
print(Employee.company)   # KSR Datavizon
```

Expected Output:

```
Acme
KSR Datavizon
KSR Datavizon
```

To change the shared value for **everyone**, assign via the class: `Employee.company = "..."`.

Visual — one shared class attribute, many instance attributes:



Key takeaways:

- Instance attrs = per object (`self.x`)
- Class attrs = shared (`Class.x`)
- `obj.x = ...` on a class attr name **shadows** for that object only
- Locals live only inside the method

Common Error: putting `students = []` at **class** level — every batch shares one list. Always create the list in `__init__`.

Note:

- Mutable class attribute → shared state across objects — initialize lists/dicts in `__init__`, not at class level.
- Prefer `ClassName.attr` when you mean the shared value (clearer than `self.attr` for class data).
- Locals are fine for temporary calculations; do not invent `self.tmp` for values you never need later.

Try these (3 project examples):

Program 1 — AcmeHR shared company name + per-employee id

```
class Employee:
    company = "AcmeHR Bengaluru"

    def __init__(self, emp_id, name, city):
        self.emp_id = emp_id
        self.name = name
        self.city = city

    def badge(self):
        return f"{self.name} ({self.emp_id}) — {Employee.company}, {self.city}"

e1 = Employee(501, "Priya Shah", "Bengaluru")
e2 = Employee(502, "Ravi Menon", "Mumbai")
print(e1.badge())
print(e2.badge())

Employee.company = "AcmeHR India"
print(e1.badge())
print(e2.badge())  # both pick up the shared class value
```

Expected Output:

```
Priya Shah (501) — AcmeHR Bengaluru, Bengaluru
Ravi Menon (502) — AcmeHR Bengaluru, Mumbai
Priya Shah (501) — AcmeHR India, Bengaluru
Ravi Menon (502) — AcmeHR India, Mumbai
```

Program 2 — TechMart shared-cart trap (spot the bug, then the fix)

```
class CartBroken:
    items = []   # WRONG — every customer shares one list
```

```
    def __init__(self, customer):
        self.customer = customer

    def add(self, sku):
        self.items.append(sku)

c1 = CartBroken("Priya")
c2 = CartBroken("Ravi")
c1.add("NB-A4")
print("Broken Ravi cart:", c2.items)  # ['NB-A4'] — leaked!

class Cart:
    def __init__(self, customer):
        self.customer = customer
        self.items = []   # CORRECT — per object

    def add(self, sku):
        self.items.append(sku)

ok1 = Cart("Priya")
ok2 = Cart("Ravi")
ok1.add("NB-A4")
print("Fixed Ravi cart:", ok2.items)  # []
```

Expected Output:

```
Broken Ravi cart: ['NB-A4']
Fixed Ravi cart: []
```

Program 3 — SecureBank shared IFSC + local fee inside a method

```
class BranchAccount:
    ifsc = "SBIN0001234"   # class — one branch code for every account here

    def __init__(self, holder, balance):
        self.holder = holder
        self.balance = float(balance)

    def transfer_out(self, amount):
        fee = 5.0   # local — exists only during this transfer
        if amount + fee > self.balance:
            print("Not enough for amount + fee")
            return
        self.balance -= amount + fee
        print(f"{self.holder} sent ₹{amount:.0f} (fee ₹{fee:.0f}) via {BranchAccount.ifsc}")

a = BranchAccount("Priya", 1000)
a.transfer_out(200)
print("Balance left:", a.balance)
# print(fee)  # NameError — local died when the method ended
```

Expected Output:

```
Priya sent ₹200 (fee ₹5) via SBIN0001234
Balance left: 795.0
```

3. Three kinds of methods (instance / class / static)

Real-world analogy: A **canteen counter**. `bill()` uses *your* tray (instance method — needs `self`). `today_menu()` is the same board for every student (class method — needs the class, not one tray). `gst(amount)` is a calculator anyone can tap without taking a token (static method — no `self` / `cls`). **TechMart** counts orders with `@classmethod`. **SecureBank** computes GST with `@staticmethod`. **MedCare** builds a `Patient` from an Aadhaar CSV row with an alternate constructor.

📦 Order | 🏦 Bank | 📱 UPI | 💰 Payroll

Where used? Line totals and deposits (instance); object counters / "build from CSV row" (class); GST / fare math helpers (static).

Definition:

- **Instance method** — first arg `self`; uses object data
- **Class method** — `@classmethod`, first arg `cls`; works with class-level data or alternate constructors
- **Static method** — `@staticmethod`; utility namespaced on the class (no `self` / `cls`)

Purpose: Not every helper needs an instance. Object counters and "build from dict/CSV" fit class methods; pure converters fit static methods.

```
class Order:
    order_count = 0

    def __init__(self, product, qty):
        self.product = product
        self.qty = qty
        Order.order_count += 1

    def line_total(self, price):          # instance method
        return self.qty * price

    @classmethod
    def get_count(cls):                   # class method
        return cls.order_count

    @classmethod
    def from_pair(cls, product, qty):     # alternate constructor
        return cls(product, qty)

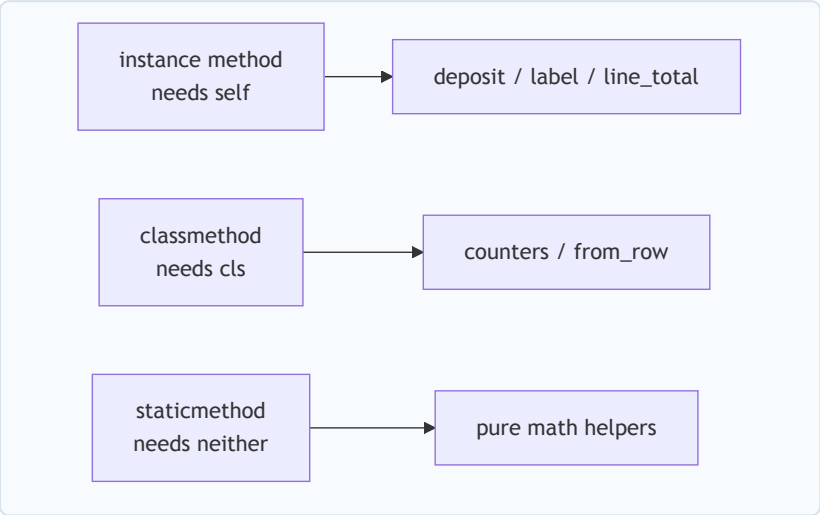
    @staticmethod
    def gst_amount(amount, rate=0.18):    # static utility
        return amount * rate

o = Order.from_pair("Notebook", 3)
print("line total:", o.line_total(40))
print("order count:", Order.get_count())
print("GST on 100:", Order.gst_amount(100))
print("product:", o.product)
```

Expected Output:

```
line total: 120
order count: 1
GST on 100: 18.0
product: Notebook
```

Visual — which method gets what?



Key takeaways:

- Instance = normal day-to-day methods
- `@classmethod` gets `cls` — class-level work / alternate constructors
- `@staticmethod` is a plain utility living on the class

Interview: `@classmethod` gets `cls` (class-level / alternate constructors). `@staticmethod` is a plain utility on the class — no `self` / `cls`.

Note:

- In day-to-day apps you mostly write **instance** methods. Use `@classmethod` / `@staticmethod` when the intent is clear — not for decoration.
- Alternate constructors (`from_pair`, `from_dict`) keep `__init__` simple and give callers a clear name for each creation path.
- Calling `Order.gst_amount(100)` or `o.gst_amount(100)` both work for static methods; prefer the class name when it is a pure utility.

Try these (3 project examples):

Program 1 — TechMart order counter + build-from-CSV

```
class Order:
    count = 0

    def __init__(self, order_id, sku, qty):
        self.order_id = order_id
        self.sku = sku
        self.qty = int(qty)
        Order.count += 1

    def label(self):
        return f"{self.order_id}: {self.sku} x{self.qty}"

    @classmethod
    def how_many(cls):
        return cls.count
```

```
    @classmethod
    def from_csv_row(cls, row):
        order_id, sku, qty = row.split(",")
        return cls(order_id.strip(), sku.strip(), qty)

o1 = Order("ORD-101", "NB-A4", 3)
o2 = Order.from_csv_row("ORD-102,PEN-01,5")
print(o1.label())
print(o2.label())
print("orders today:", Order.how_many())
```

Expected Output:

```
ORD-101: NB-A4 x3
ORD-102: PEN-01 x5
orders today: 2
```

Program 2 — SecureBank GST / UPI fee (static utilities)

```
class BankMath:
    @staticmethod
    def with_gst(amount, rate=0.18):
        return amount * (1 + rate)

    @staticmethod
    def upi_fee(amount):
        if amount <= 0:
            return 0.0
        return 0 if amount < 1000 else 2.0

print("₹1000 + GST:", round(BankMath.with_gst(1000), 2))
print("UPI fee on ₹200:", BankMath.upi_fee(200))
print("UPI fee on ₹2500:", BankMath.upi_fee(2500))
```

Expected Output:

```
₹1000 + GST: 1180.0
UPI fee on ₹200: 0
UPI fee on ₹2500: 2.0
```

Program 3 — MedCare Patient from Aadhaar CSV row

```
class Patient:
    def __init__(self, name, age, ward):
        self.name = name
        self.age = int(age)
        self.ward = ward

    def card(self):
        return f"{self.name}, {self.age}y, ward {self.ward}"

    @classmethod
    def from_csv_row(cls, row):
        name, age, ward = row.split(",")
        return cls(name.strip(), age.strip(), ward.strip())

p1 = Patient("Anita Sharma", 28, "OPD")
```

```
p2 = Patient.from_csv_row("Ravi Menon, 41, ICU")
print(p1.card())
print(p2.card())
```

Expected Output:

```
Anita Sharma, 28y, ward OPD
Ravi Menon, 41y, ward ICU
```

---

4. Encapsulation — definition, problem, then `_` / `__` / `@property`

Real-world analogy: You cannot open an **ATM vault** with a screwdriver. You press **Withdraw**. The machine checks PIN and balance, then dispenses cash. Encapsulation is that locked vault plus the buttons. If anyone could type `acct.balance = -50000`, **SecureBank** would be bankrupt on paper. **TechMart** stock cannot go negative because a tired intern edited a field. **MedCare** vitals cannot be overwritten with `"N/A"` from a random script.

🏦 Bank 📱 UPI

Where used? SecureBank balances, UPI wallets, passwords, **TechMart** inventory stock — anything with a rule that must stay true after every change.

Definition (say this in class)

Encapsulation means two things together:

1. **Bundle** related data and the methods that use that data into one class (the object "owns" its state).
2. **Hide and control** that state — outside code should not poke the raw fields; it must go through a **public API** (`deposit`, `withdraw`, `get_balance` / `@property`) that enforces the rules.

Interview one-liner: *"Encapsulation = data + behaviour in one place, with controlled access so invariants cannot be broken from outside."*

It is **not** "putting underscores on variables." Underscores and `@property` are *tools*. The *goal* is protecting the object's rules.

Exact problem it solves

Without encapsulation, the object's important data is **naked**. Any line of code anywhere can assign nonsense and break the business rule — and the class never gets a chance to say "no".

Business rule we care about (bank example):
`balance` must never become negative by a casual assignment; money may change only through validated `deposit` / `withdraw`.

Broken — no encapsulation (the bug factory):

```
class OpenAccount:
    def __init__(self, holder, balance):
        self.holder = holder
        self.balance = balance   # public raw field

    def withdraw(self, amount):
        if amount > self.balance:
            print("Insufficient funds")
```

```
            return
        self.balance -= amount

acct = OpenAccount("Priya", 1000)
acct.withdraw(200)          # OK — uses the method
print("after withdraw:", acct.balance)

# Later… a tired teammate (or your own shortcut) skips the method:
acct.balance = -50000       # accepted! rule destroyed
acct.balance = "N/A"        # accepted! type destroyed
print("corrupted balance:", acct.balance)
```

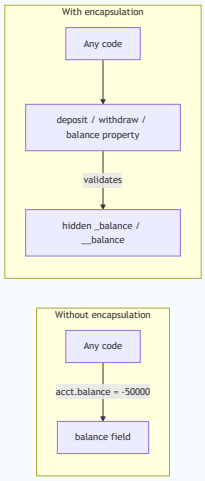
Expected Output:

```
after withdraw: 800
corrupted balance: N/A
```

Problems this causes in real projects:

What goes wrong	Why it hurts
Outside code writes `acct.balance = …`	Rules in `withdraw` are **bypassed**
Invalid values sneak in (`-50000` , `"N/A"`)	Downstream math / UIs / reports crash or lie
Fixing the rule means hunting **every** assignment	Same pain as copy-paste functions — scattered writes
You cannot change how balance is stored later	Callers are glued to the raw field name

Visual — naked field vs guarded API:



Key takeaways:

- Exact problem: **uncontrolled writes** to important state
- Method rules are useless if callers can skip them and edit the field
- Encapsulation forces “change money only through the door we control”

Fixed direction (idea first — syntax next):

- Keep the real value **inside** (`_balance` or `__balance`)
- Expose only safe operations: `deposit` , `withdraw` , and a read via `get_balance()` or `@property`
- Callers who follow the API cannot casually invent `-50000`

That is the problem encapsulation solves. Naming (`_` / `__`) and `@property` are how Python helps you express that design.

Teach the tools in **three layers**:

1. Naming: public vs `_name` vs `__name`
2. Safe access: `get_balance()` **or** `@property`
3. Optional controlled write: `@x.setter`

4a. Public vs `_name` vs `__name` (naming)

Syntax	Name people use	What Python does	What it means for you
`name`	**public**	Normal attribute	Part of the API — callers may read/write
`_name`	**protected** (by convention)	Still a normal attribute	"Internal — please don't touch from outside." Python does **not** block access
`__name`	**name-mangled** (often called private in notes)	Stored as `_ClassName__name`	Accidental `obj.__name` fails; determined access can still use the mangled name

Important: None of these is encryption or real security. Encapsulation is for **honest, clear code** — teammates follow the public API.

Side-by-side demo — `_balance` vs `__balance`:

```
class AccountUnderscore:
    def __init__(self, balance):
        self._balance = float(balance)   # convention only

class AccountDunder:
    def __init__(self, balance):
        self.__balance = float(balance)  # name-mangled

u = AccountUnderscore(1000)
d = AccountDunder(1000)

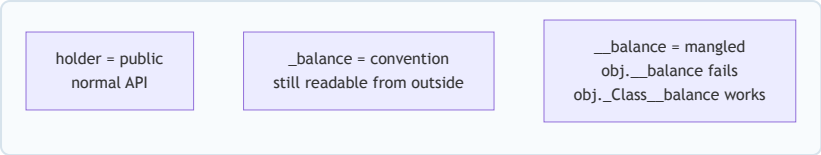
print("public-style read of _balance:", u._balance)   # works — convention only
u._balance = -50                                      # also works — dangerous
print("after bad write:", u._balance)

print("has '__balance' on d?", hasattr(d, "__balance"))           # False
# print(d.__balance)   # do not uncomment — crashes (name is mangled)
print("mangled name still exists:", d._AccountDunder__balance)    # works if you insist
```

Expected Output:

```
public-style read of _balance: 1000.0
after bad write: -50.0
has '__balance' on d? False
mangled name still exists: 1000.0
```

Visual — three access styles:



Key takeaways:

- `_balance` = polite “hands off” — Python does **not** stop you
- `__balance` = renamed to `_ClassName__balance` — stops *accidental* access
- Neither is true privacy / security
- Prefer changing money only through `deposit` / `withdraw`

Which should students use?

Style	When	Course note
`_balance` + `@property`	Idiomatic modern Python	Preferred in notes / interviews
`__balance` + `get_balance()`	Clear “don’t touch” + Java-like getter	**Lab kit** style — both OK

Common Error: saying “ `__balance` is private so nobody can read it.” They can: `obj._BankAccount__balance`.

4b. `@property` — attribute-like read that runs a method

Problem with a raw public attribute:

```
class BadAccount:
    def __init__(self, balance):
        self.balance = balance

acct = BadAccount(1000)
acct.balance = -999999   # no rule — nonsense accepted
print(acct.balance)
```

Expected Output:

```
-999999
```

Fix idea: keep real money in `_balance` (or `__balance`), and expose a **safe read**.

Two equivalent public APIs:

Style	Call site	Internals
Getter method (lab style)	`acct.get_balance()`	often `__balance`
`@property` (idiomatic)	`acct.balance`	often `_balance`

`@property` turns a method into something you **read like an attribute** (no parentheses):

```
class BankAccount:
    def __init__(self, holder, balance=0):
        self.holder = holder          # public — OK to read
        self._balance = float(balance)  # internal storage

    @property
    def balance(self):                # looks like attribute, runs method
        return self._balance

    def deposit(self, amount):
        if amount <= 0:
            print("Invalid deposit amount")
            return
        self._balance += amount

    def withdraw(self, amount):
        if amount <= 0:
            print("Invalid withdraw amount")
            return
        if amount > self._balance:
            print("Insufficient funds")
            return
        self._balance -= amount

acct = BankAccount("Priya", 1000)
acct.deposit(500)
acct.withdraw(200)
acct.withdraw(5000)                 # rejected
print("holder:", acct.holder)
print("balance via @property:", acct.balance)   # no ()
# acct.balance = 999999   # do not uncomment — crashes (read-only; no setter yet)
```

Expected Output:

```
Insufficient funds
holder: Priya
balance via @property: 1300.0
```

Same idea with lab-style getter (`__balance`):

```
class BankAccountLabStyle:
    def __init__(self, holder, balance=0):
        self.holder = holder
        self.__balance = float(balance)

    def get_balance(self):            # must call with ()
        return self.__balance

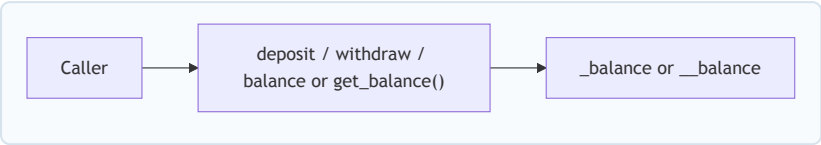
    def deposit(self, amount):
        if amount > 0:
            self.__balance += amount

lab = BankAccountLabStyle("Anita", 1000)
lab.deposit(200)
print("balance via getter:", lab.get_balance())
```

Expected Output:

```
balance via getter: 1200.0
```

Visual — outside talks to API; inside holds real state:



Key takeaways:

- `@property` = read with `obj.balance` (no `()`)
- Getter = read with `obj.get_balance()` (needs `()`)
- Real money stays in `_balance` / `__balance`
- Without a setter, `obj.balance = ...` fails — force changes through methods

Remember: `@property` does **not** replace encapsulation by itself. You still store the real value in `_…` / `__…` and validate in `deposit` / `withdraw`.

4c. Optional — `@balance.setter` (controlled write)

Use a setter only when assignment should be allowed **with rules**. For bank money, prefer `deposit` / `withdraw`. Setter fits things like renaming with validation:

```
class Student:
    def __init__(self, name):
        self._name = name

    @property
    def name(self):
        return self._name

    @name.setter
    def name(self, value):
        value = value.strip()
        if not value:
            print("Name cannot be empty")
            return
        self._name = value

s = Student("Anita")
print(s.name)
s.name = "  Anita Sharma  "
print(s.name)
s.name = "   "            # rejected
print("still:", s.name)
```

Expected Output:

```
Anita
Anita Sharma
Name cannot be empty
still: Anita Sharma
```

Note:

- Leading with `__` on every field is noisy. Start with public + validated methods; add `_` / `__` / `@property` when the API needs it.
- Interview line: "`_name` is convention; `__name` is name mangling; `@property` gives a clean read API over internal storage."
- Course rule of thumb: **lab kit** may use `__balance` + `get_balance()`; **notes / interviews** prefer `_balance` + `@property`.

Lab vs note style: Lab kit = `__balance` + `get_balance()`. Notes above = `_balance` + `@property`. Same goal — both acceptable in this course.

Common Error: treating `__balance` as encryption / security. It only renames the attribute (`_BankAccount__balance`).

Try these (3 project examples):

Program 1 — SecureBank UPIWallet with `__balance` + getter (lab style)

```
class UPIWallet:
    def __init__(self, balance=0):
        self.__balance = float(balance)

    def credit(self, amount):
        if amount <= 0:
            print("Invalid credit amount")
            return
        self.__balance += amount

    def debit(self, amount):
        if amount <= 0:
            print("Invalid debit amount")
            return
        if amount > self.__balance:
            print("Insufficient balance")
            return
        self.__balance -= amount

    def get_balance(self):
        return self.__balance

wallet = UPIWallet(500)
wallet.credit(200)
wallet.debit(100)
wallet.debit(1000)  # rejected
print("Balance:", wallet.get_balance())
print("Direct '__balance' visible?", hasattr(wallet, "__balance"))  # False
```

```
# print(wallet.__balance)  # do not uncomment — crashes; use get_balance()
print("Mangled peek (don't do this in apps):", wallet._UPIWallet__balance)
```

Expected Output:

```
Insufficient balance
Balance: 600.0
Direct '__balance' visible? False
Mangled peek (don't do this in apps): 600.0
```

Program 2 — MedCare vitals: `@property` read-only + computed band

```
class Vitals:
    def __init__(self, celsius):
        self._celsius = float(celsius)   # internal

    @property
    def celsius(self):                   # read-only public view
        return self._celsius

    @property
    def fahrenheit(self):                # computed — no stored F
        return self._celsius * 9 / 5 + 32

    @property
    def fever(self):
        return self._celsius >= 38.0

v = Vitals(37)
print("C:", v.celsius)
print("F:", round(v.fahrenheit, 1))
print("fever?", v.fever)
# v.celsius = 40   # do not uncomment — crashes (read-only property; no setter)
print("celsius is read-only — change only via a setter or a method (later design)")
```

Expected Output:

```
C: 37.0
F: 98.6
fever? False
celsius is read-only — change only via a setter or a method (later design)
```

Program 3 — TechMart stock cannot go negative (public API only)

```
class Product:
    def __init__(self, sku, stock):
        self.sku = sku
        self.__stock = int(stock)

    def sell(self, qty):
        if qty <= 0:
            print("Invalid qty")
            return
        if qty > self.__stock:
            print(f"{self.sku}: not enough stock")
            return
        self.__stock -= qty
```

```
    def restock(self, qty):
        if qty <= 0:
            print("Invalid qty")
            return
        self.__stock += qty

    def get_stock(self):
        return self.__stock

p = Product("NB-A4", 10)
p.sell(3)
p.sell(20)   # rejected
p.restock(5)
print("Stock:", p.get_stock())
print("Direct '__stock' visible?", hasattr(p, "__stock"))
```

Expected Output:

```
NB-A4: not enough stock
Stock: 12
Direct '__stock' visible? False
```

Step-by-Step Annotated Python 3 Code

Examples below deepen Part 1 ideas. Type them; predict the output; then run.

Example 1 — BankAccount with encapsulation (lab-shaped)

This mirrors the lab kit style (`__balance` + `get_balance`).

```
class BankAccount:
    def __init__(self, holder, balance=0):
        self.holder = holder
        self.__balance = float(balance)

    def deposit(self, amount):
        if amount <= 0:
            print("Invalid deposit amount")
            return
        self.__balance += amount
        print(f"Deposited ₹{amount}. New balance: ₹{self.__balance}")

    def withdraw(self, amount):
        if amount <= 0:
            print("Invalid withdraw amount")
            return
        if amount > self.__balance:
            print("Insufficient funds")
            return
        self.__balance -= amount
        print(f"Withdrew ₹{amount}. New balance: ₹{self.__balance}")

    def get_balance(self):
        return self.__balance

acct = BankAccount("Priya", 1000)
acct.deposit(500)
```

```
acct.withdraw(200)
acct.withdraw(-50)     # rejected
acct.deposit(0)        # rejected
print("Balance:", acct.get_balance())
print("Holder:", acct.holder)
# print(acct.__balance)  # do not uncomment — crashes (name mangling); use get_balance()
```

Expected Output:

```
Deposited ₹500. New balance: ₹1500.0
Withdrew ₹200. New balance: ₹1300.0
Invalid withdraw amount
Invalid deposit amount
Balance: 1300.0
Holder: Priya
```

Remember: validation lives in methods; callers never set `__balance` directly.

Example 2 — Full Student (marks list as instance data)

```
class Student:
    college = "KSR College"

    def __init__(self, roll, name, marks):
        self.roll = roll
        self.name = name
        self.marks = list(marks)   # own copy

    def average(self):
        if not self.marks:
            return 0.0
        return sum(self.marks) / len(self.marks)

    def grade(self):
        avg = self.average()
        if avg >= 85:
            return "A"
        if avg >= 70:
            return "B"
        if avg >= 50:
            return "C"
        return "Fail"

    def add_mark(self, mark):
        if mark < 0 or mark > 100:
            print("Mark must be 0–100")
            return
        self.marks.append(mark)

    def report(self):
        print(f"{self.roll} | {self.name} | {Student.college}")
        print(f"  marks={self.marks} avg={self.average():.1f} grade={self.grade()}")

s = Student(101, "Anita", [72, 68, 81])
s.add_mark(90)
s.add_mark(150)   # rejected
s.report()
```

Expected Output:

```
Mark must be 0–100
101 | Anita | KSR College
  marks=[72, 68, 81, 90] avg=77.8 grade=B
```

Example 3 — Product catalogue with classmethod + static GST

```
class Product:
    catalogue_count = 0

    def __init__(self, sku, name, price):
        self.sku = sku
        self.name = name
        self.price = float(price)
        Product.catalogue_count += 1

    def label(self):
        return f"{self.sku}: {self.name} @ ₹{self.price:.0f}"

    def price_with_gst(self):
        return self.price + Product.gst_amount(self.price)

    @classmethod
    def how_many(cls):
        return cls.catalogue_count

    @classmethod
    def from_dict(cls, data):
        return cls(data["sku"], data["name"], data["price"])

    @staticmethod
    def gst_amount(amount, rate=0.18):
        return amount * rate

p1 = Product("PEN-01", "Blue Pen", 10)
p2 = Product.from_dict({"sku": "NB-A4", "name": "Notebook", "price": 40})
print(p1.label())
print(p2.label())
print("GST on pen:", Product.gst_amount(p1.price))
print("Pen with GST:", p1.price_with_gst())
print("products created:", Product.how_many())
```

Expected Output:

```
PEN-01: Blue Pen @ ₹10
NB-A4: Notebook @ ₹40
GST on pen: 1.8
Pen with GST: 11.8
products created: 2
```

Practice Exercises

(Each Practice item includes a filled Model answer and Expected Output where the demo prints.)

Basic

1. Create `Laptop` with `brand`, `ram_gb`, and a `specs()` method that prints both.
   Model answer:

```
class Laptop:
    def __init__(self, brand, ram_gb):
        self.brand = brand
        self.ram_gb = ram_gb

    def specs(self):
        print(f"{self.brand} — {self.ram_gb} GB RAM")

Laptop("Dell", 16).specs()
```

Expected Output:

```
Dell — 16 GB RAM
```

1. Create two `Laptop` objects and call `specs()` on each.
   Model answer:

```
l1 = Laptop("Dell", 16)
l2 = Laptop("Lenovo", 8)
l1.specs()
l2.specs()
```

Expected Output:

```
Dell — 16 GB RAM
Lenovo — 8 GB RAM
```

1. Add `upgrade_ram(extra_gb)` that increases `ram_gb` only when `extra_gb > 0`.
   Model answer:

```
class Laptop:
    def __init__(self, brand, ram_gb):
        self.brand = brand
        self.ram_gb = ram_gb

    def upgrade_ram(self, extra_gb):
        if extra_gb <= 0:
            print("extra_gb must be positive")
            return
        self.ram_gb += extra_gb

    def specs(self):
        print(f"{self.brand} — {self.ram_gb} GB RAM")

lap = Laptop("Dell", 8)
lap.upgrade_ram(8)
lap.upgrade_ram(-2)
lap.specs()
```

Expected Output:

```
extra_gb must be positive
Dell — 16 GB RAM
```

1. Create `Product` with `sku`, `price`, and `display()` (do **not** use `__str__` yet — that is Part 3).
   Model answer:

```
class Product:
    def __init__(self, sku, price):
        self.sku = sku
        self.price = price

    def display(self):
        print(f"{self.sku} @ ₹{self.price}")

Product("PEN-01", 10).display()
```

Expected Output:

```
PEN-01 @ ₹10
```

1. Add `apply_discount(percent)` returning the discounted price (do not mutate unless you choose to).
   Model answer:

```
class Product:
    def __init__(self, sku, price):
        self.sku, self.price = sku, price

    def apply_discount(self, percent):
        return self.price * (1 - percent / 100)

print(Product("X", 1000).apply_discount(10))  # 900.0
```

Expected Output:

```
900.0
```

1. Create `Employee` with a **class** attribute `company = "KSR Datavizon"` and instance attributes `emp_id`, `name`. Print both employees after changing `Employee.company`.
   Model answer:

```
class Employee:
    company = "KSR Datavizon"

    def __init__(self, emp_id, name):
        self.emp_id = emp_id
        self.name = name

    def card(self):
        print(f"{self.emp_id}: {self.name} @ {Employee.company}")
```

```
e1 = Employee(1, "Ravi")
e2 = Employee(2, "Meera")
Employee.company = "KSR Labs"
e1.card()
e2.card()
```

Expected Output:

```
1: Ravi @ KSR Labs
2: Meera @ KSR Labs
```

1. Build a tiny `Wallet` with `_balance`, `@property` `balance`, and `deposit` only (no direct write to balance).
   Model answer:

```
class Wallet:
    def __init__(self, opening=0):
        self._balance = float(opening)

    @property
    def balance(self):
        return self._balance

    def deposit(self, amount):
        if amount > 0:
            self._balance += amount

w = Wallet(100)
w.deposit(50)
print(w.balance)  # 150.0
```

Expected Output:

```
150.0
```

1. Lab warm-up: write `BankAccount` with `__balance`, `deposit`, `withdraw`, `get_balance` — reject amount ≤ 0 and overdraft. Then open the lab kit and finish it properly.
   Model answer:

```
class BankAccount:
    def __init__(self, holder, balance=0):
        self.holder = holder
        self.__balance = float(balance)

    def deposit(self, amount):
        if amount <= 0:
            print("Invalid deposit amount")
            return
        self.__balance += amount

    def withdraw(self, amount):
        if amount <= 0:
            print("Invalid withdraw amount")
            return
        if amount > self.__balance:
            print("Insufficient funds")
```

```
            return
        self.__balance -= amount

    def get_balance(self):
        return self.__balance

acct = BankAccount("Priya", 1000)
acct.deposit(500)
acct.withdraw(200)
acct.withdraw(5000)
print("Balance:", acct.get_balance())
```

Expected Output:

```
Insufficient funds
Balance: 1300.0
```

Next

Assignment reminder: portfolio push is **Mini-Project (Assignment): BankAccount OOP CLI** in Part 3. Keep building the lab kit.

Continue with **OOP Part 2 — Inheritance & Composition**.